

GYAN MITRA HACKATHON 2026

Team Name: Griffins

Team Leader Name: Agiless B B

Theme: Intelligent Healthcare

Idea Title: HealthLink - Secure, Local

GenAI-Powered Personal Health Knowledge Graph.

Institution name: Saranathan College of Engineering

Problem Statement

- **The Conflict:** Health data is trapped in "Dark Silos"—wearables (lifestyle) vs. hospital portals (clinical).
- **The Impact:** This fragmentation prevents doctors from seeing a complete patient history, especially during critical "Day-1" first visits.
- **The Risk:** Current cloud-based health platforms pose severe privacy risks and provide generic, non-personalized advice.
- **Economic Cost:** Fragmentation leads to inefficient consultations, delayed clinical decisions, and preventable complications.

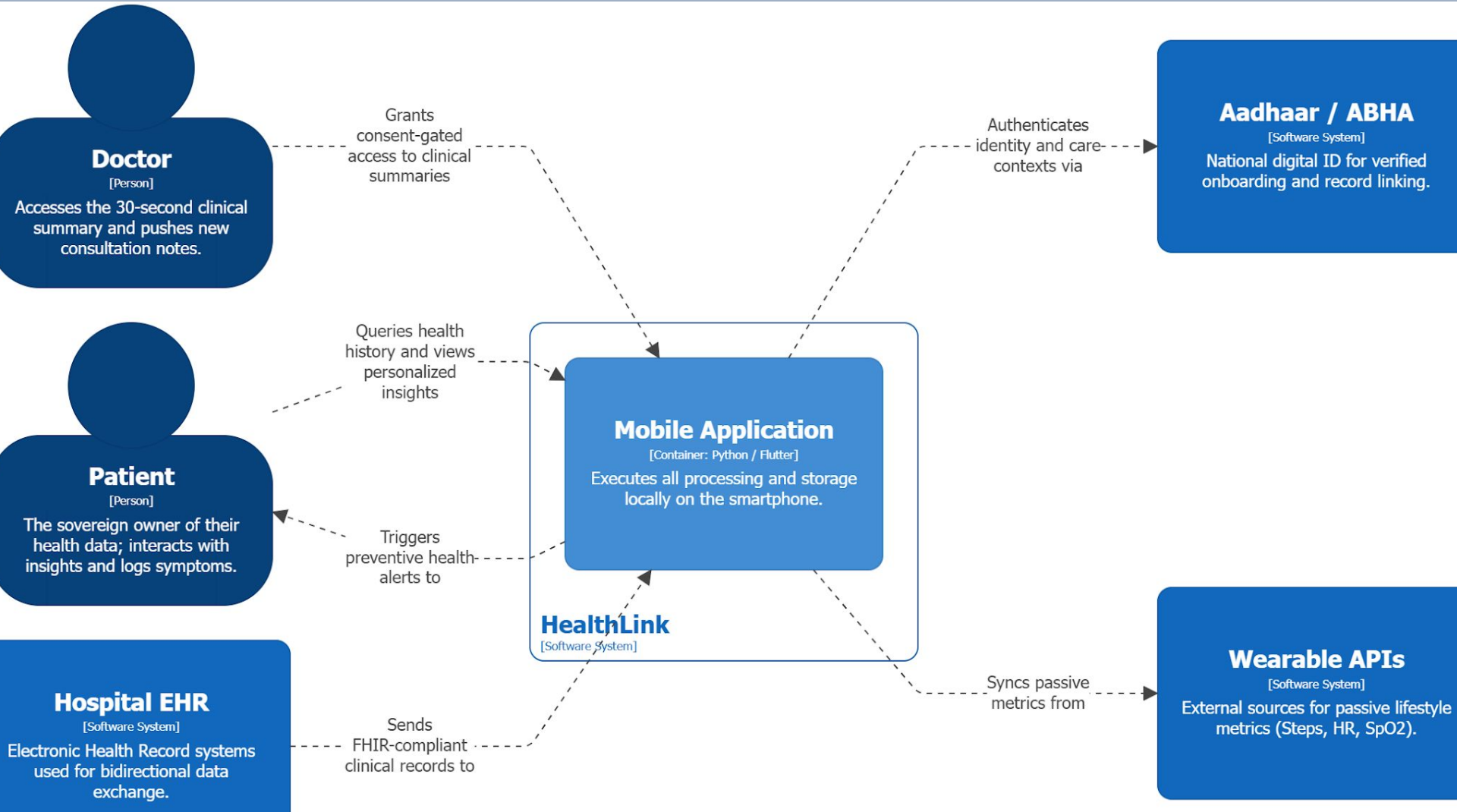
Proposed Solution

HealthLink is a decentralized, privacy-first platform that transforms fragmented medical data into a unified **Semantic Knowledge Graph (KG)** residing entirely on the user's device. By merging wearable streams with clinical records, it creates a longitudinal "health thread" accessible via local, quantized GenAI for secure, grounded medical insights.

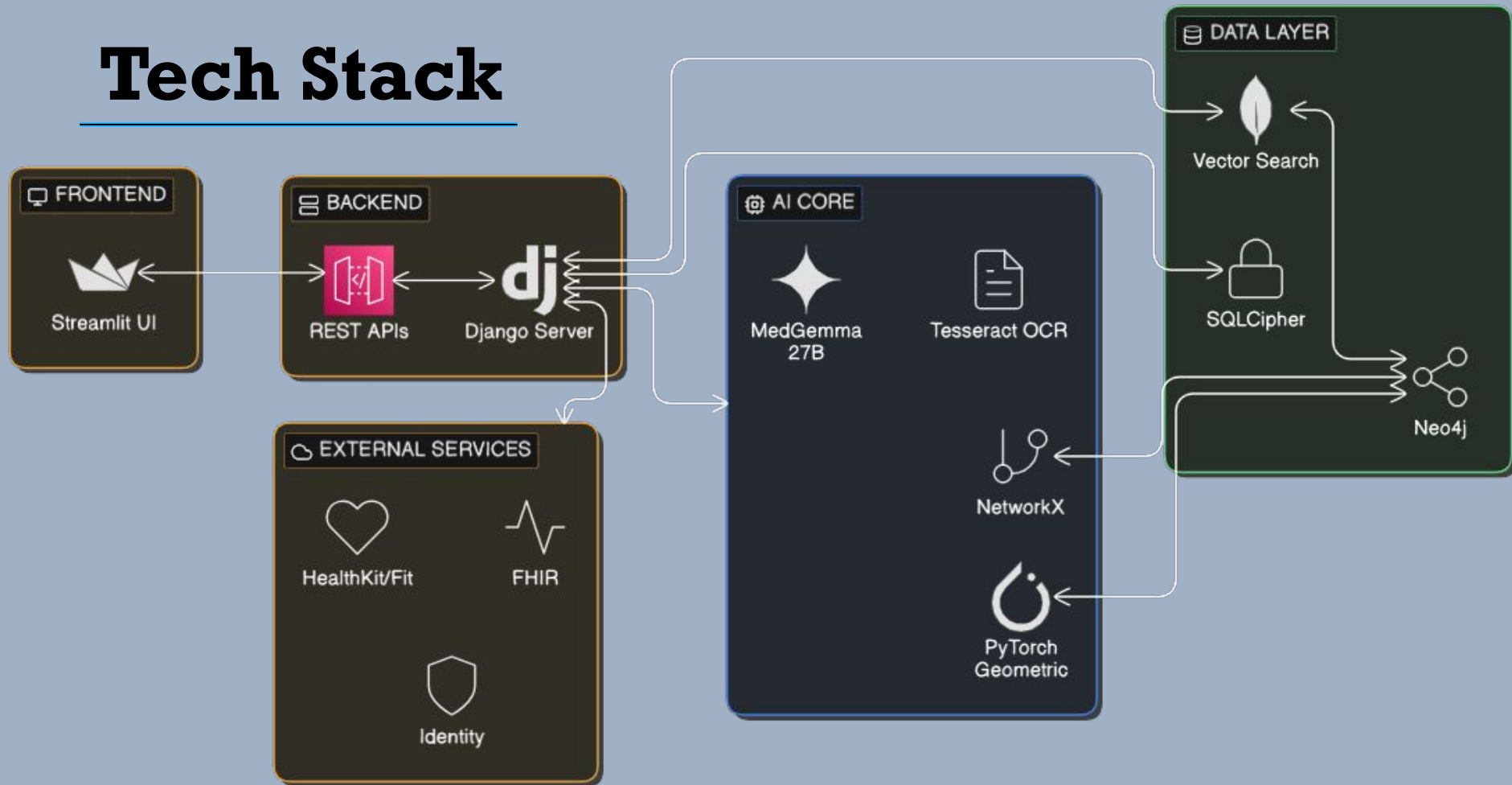
The 5-Step Workflow

- 1.Sovereign Onboarding:** Aadhaar-verified account creation and initialization of a secure local user database.
- 2.Multimodal Ingestion:** Automated capture of wearable APIs (Apple/Google) and on-device OCR for PDF/CSV clinical reports.
- 3.GraphRAG Synthesis:** Automated extraction of entities and relationships to link lifestyle metrics (e.g., activity) to clinical markers (e.g., HbA1c).
- 4.Local Hybrid RAG:** A local, quantized LLM traverses the KG and FAISS vector search to provide 100% grounded, natural-language answers.
- 5.Bidirectional Continuity:** Consent-gated "Write-Back" where hospitals add consultation notes and lab results directly to the user's KG via FHIR JSON standards.

Block Diagram



Tech Stack



- **Frontend & Backend:** Streamlit (UI) and Django (Core Orchestration).
- **Databases:** MongoDB Atlas (Hybrid Vector Search) and SQLCipher (Encrypted Local Relational DB). Neo4j (Graph Database)
- **AI/ML Frameworks:** NetworkX (Graph Management), and PyTorch Geometric (GNN Anomaly Detection).
- **Models:** Quantized Llama-3 8B / MedGemma 27B (On-device LLMs) and Tesseract OCR.
- **Interoperability:** FHIR R4/R5 standards and Aadhaar/ABHA Sandbox APIs.

Innovation

- **Semantic Data Fusion (GraphRAG)**
 - **Beyond Lockers:** Doesn't just store files; it "understands" how **lifestyle** (steps/sleep) correlates with **clinical** (HbA1c) outcomes.
 - **Multi-Hop Reasoning:** Maps complex connections across years of fragmented history to uncover hidden health trends.
- **Zero-Leak Edge AI**
 - **On-Device Inference:** Quantized GenAI runs 100% locally on your device sensitive medical data **never** leaves your control.
 - **Absolute Privacy:** Eliminates cloud-sync vulnerabilities while providing high-end clinical insights.
- **Bidirectional Continuity**
 - **Verified Write-Back:** First system allowing hospitals to securely push notes and lab results back into the **user's own** Knowledge Graph.
 - **Unified History:** Bridges the gap between disparate clinics for a truly lifelong, portable health record.
- **Predictive GNN Alerts**
 - **Proactive Care:** Uses **Graph Neural Networks** to detect subtle physiological anomalies (like arrhythmia) before they become critical.
 - **Smart Baselines:** Learns your personal "normal" instead of relying on generic medical averages.

Economic and Social Impacts

- **Economic Impact (The Financial Value)**

- **Operational Efficiency for Hospitals:** The **30-Second Clinical Summary** reduces the "History Taking" phase of a consultation by **~40%**. This allows government OPDs to handle higher patient volumes with better diagnostic accuracy.

- **Early Intervention Savings:** By using **GNN-based anomaly detection**, we move from "Emergency Care" (expensive) to "Preventive Care" (affordable). Detecting a heart condition early is **10x cheaper** than treating a cardiac arrest.

- **Social Impact (The Human Value)**

- **Eliminating the "Medical Memory" Burden:** Patients no longer need to remember exact medication names, dosages, or past surgery dates. The app acts as a **Digital Health Twin** that provides instant, accurate recall during pharmacy visits or sudden doctor consultations.

- **Reducing "Medical Anxiety":** The GenAI provides **contextual explanations** of complex lab reports in simple language. Instead of Googling symptoms and panicking, users get grounded, personalized explanations of what their "High Creatinine" or "Low Vitamin D" actually means for *them*.

Commercial Viability

Explosive Indian Digital Health Growth

- Market: ~USD 14.5–18B in 2024–2025 → projected USD 107B by 2033 (CAGR 25%+)
- Driven by Ayushman Bharat Digital Mission (ABDM): 500M+ ABHA IDs created, but low real engagement due to fragmentation

Chronic Disease Epidemic

- 77M diabetics + 220M hypertensives → urgent need for preventive, personalized insights
- 80% healthcare spending after disease progression → high out-of-pocket burden on families

Wearable & Tech Boom

- 100M+ Indians with fitness trackers/smartwatches by 2026
- Privacy crisis in cloud platforms → demand for local, sovereign solutions (DPDP Act alignment)

Untapped Niche

- No privacy-first, on-device tool unifying wearable lifestyle + clinical data with bidirectional hospital flow
- TAM: Preventive digital health subset of \$100B+ market by 2030

Business Model

Freemium for Consumers (Primary Users)

- Basic: Free KG creation, simple queries, basic alerts
- Premium: Advanced insights, voice queries, proactive anomaly detection, personalized recommendations.

B2B for Hospitals & Clinics

- Nominal fees for premium tools: Rapid KG summaries for Day-1 patients, advanced analytics, FHIR integration
- Licensing/subscription model for cross-provider continuity

Government & Grant Alignment

- Government aligned project which provides this service in all hospitals across India.

Reference

- **Research Papers:**

- *From Local to Global: A Graph RAG Approach to Query-Focused Summarization* (<https://arxiv.org/abs/2404.16130>)
- *Graph-Based Deep Learning for Medical Diagnosis and Analysis: Past, Present and Future* (<https://pmc.ncbi.nlm.nih.gov/articles/PMC8309939/>)

- **Datasets:**

- *The Clinical Powerhouse: MIMIC-IV*
(<https://physionet.org/content/mimiciv/3.1/>)
- *The Wearable Specialist: BIG IDEAs Lab*
(<https://physionet.org/content/big-ideas-glycemic-wearable/1.1.2/>)